

Requirement Specification & Design Document for the Library Management System

I. Functional Requirements

Login: The system shall allow members and administrators to log in using their username and password.

- + The user enters a username and password.
- + The system verifies the login information against the stored account information.
- + If the information is correct, the system identifies the user's role and allows access to the appropriate functions.
- + If the information is incorrect, the system displays an error message.

- View Personal Information: The system shall allow members to view their personal information.

- + The displayed information includes:
 - Full name
 - Member ID
 - Email
 - Date of birth
 - Address
 - Membership type
- + The personal information is view-only. Members cannot edit their personal information through the system.
 - The system shall display the information belonging to the currently logged-in member.

- Search Books: The system shall allow members to search for books in the library.

- + The member enters a keyword such as book title or author.
- + The system searches the stored book information and displays matching books.
- + The system shall display the book title, author, and availability information.
- + If no matching book is found, the system shall display an appropriate message.

- View Borrowed Books: The system shall allow members to view their borrowed books.

- + The borrowing list shall display:
 - Book title
 - Author
 - Borrowing date

- Due date
- Borrowing status
- The system shall only display borrowing records belonging to the currently logged-in member.

- **View Fines:** The system shall allow members to view fines for overdue books.

+ The system shall:

+ Display the fine for each overdue book.

+ Display the number of overdue days.

+ Display the total outstanding fine.

+ If the member has no outstanding fines, the system shall display an appropriate message.

- **Member Management:** The system shall allow administrators to manage library members.

Administrators can:

+ View member information.

+ Add new members by entering the required member information.

+ Update existing member information.

+ Delete member accounts.

+ The system shall check the member information before saving changes.

- **Book Management:** The system shall allow administrators to manage books in the library.

Administrators can:

+ Add new books.

+ View book information.

+ Update book information.

+ Delete books.

+ Update book quantity and availability.

+ The system shall update the available quantity when book information is changed.

- **Borrow/Return Management:** The system shall allow administrators to manage book borrowing and returning.

Administrators can:

+ Record a book borrowing transaction.

+ Record a book return.

- + Set the due date for borrowed books.
- + Update book availability after borrowing or returning.
- + Check whether a returned book is overdue.
- + If a returned book is overdue, the system shall calculate and record the corresponding fine.

II. Non-Functional Requirements

- Usability: The system shall provide a simple and clear interface that is easy for members and administrators to use.

- + The system shall provide clear menus and labels for each function.
- + The system shall display clear messages when an operation is successful or fails.

- Reliability: The system shall operate stably during normal library operations.

The system shall handle errors and exceptions without causing the program to crash.

- + The system shall validate user input before processing important operations.

- Performance: The system shall process common operations such as searching books, viewing borrowed books, and managing borrowing and returning within a few seconds under normal operating conditions.

- + Book search and information retrieval should normally be completed within **3 seconds**.

- Security: The system shall require users to log in using a valid username and password.

- + The system shall control access based on user roles:
- + Members can only access their own personal information, borrowed books, and fines.
- + Administrators can access member management, book management, and borrow/return management.
- + Users shall not be allowed to access functions that are not permitted for their role.

- Data Integrity: The system shall maintain accurate and consistent information about members, books, borrowing records, and fines.

- + The system shall correctly update book availability when a book is borrowed or returned.
- + The available quantity of a book shall not be less than zero.
- + Borrowing and returning information shall be associated with the correct member and book.

- Supportability: The source code shall be organized clearly and be easy to maintain and debug.

- + The system shall separate major functions into appropriate modules or functions.

+ The source code shall use clear and consistent naming.

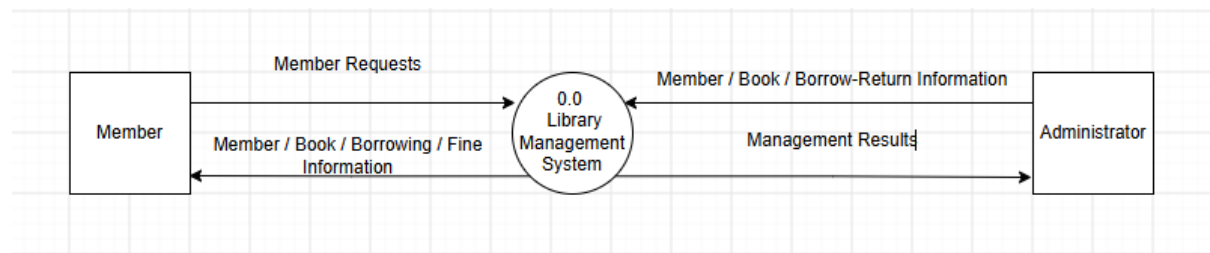
- **Design Constraints:** The system shall be developed and run directly using Python.

+ The system shall use Python as the main programming language.

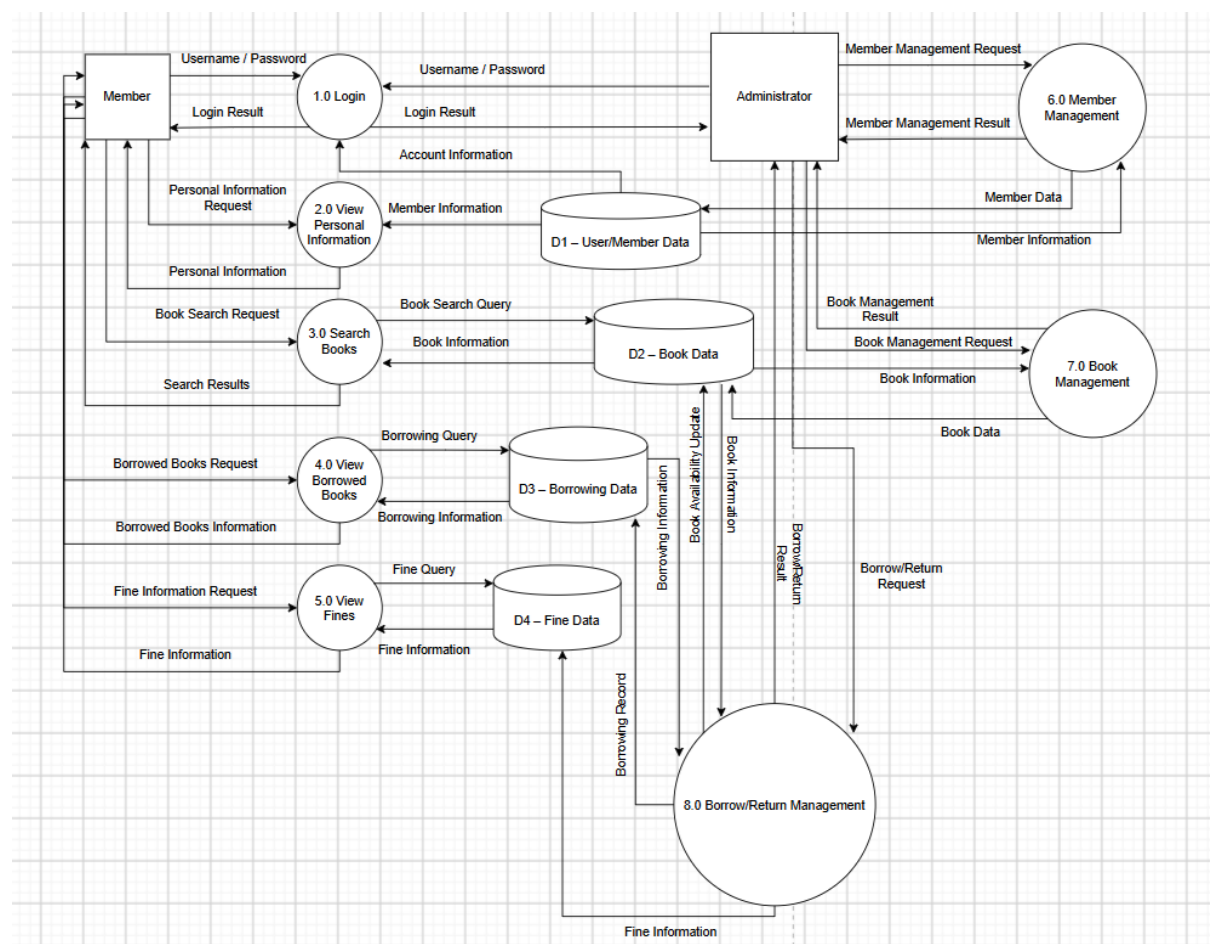
+ The system shall use an appropriate database to store member, book, borrowing, and fine information.

III. Data Flow Diagram

Level 0

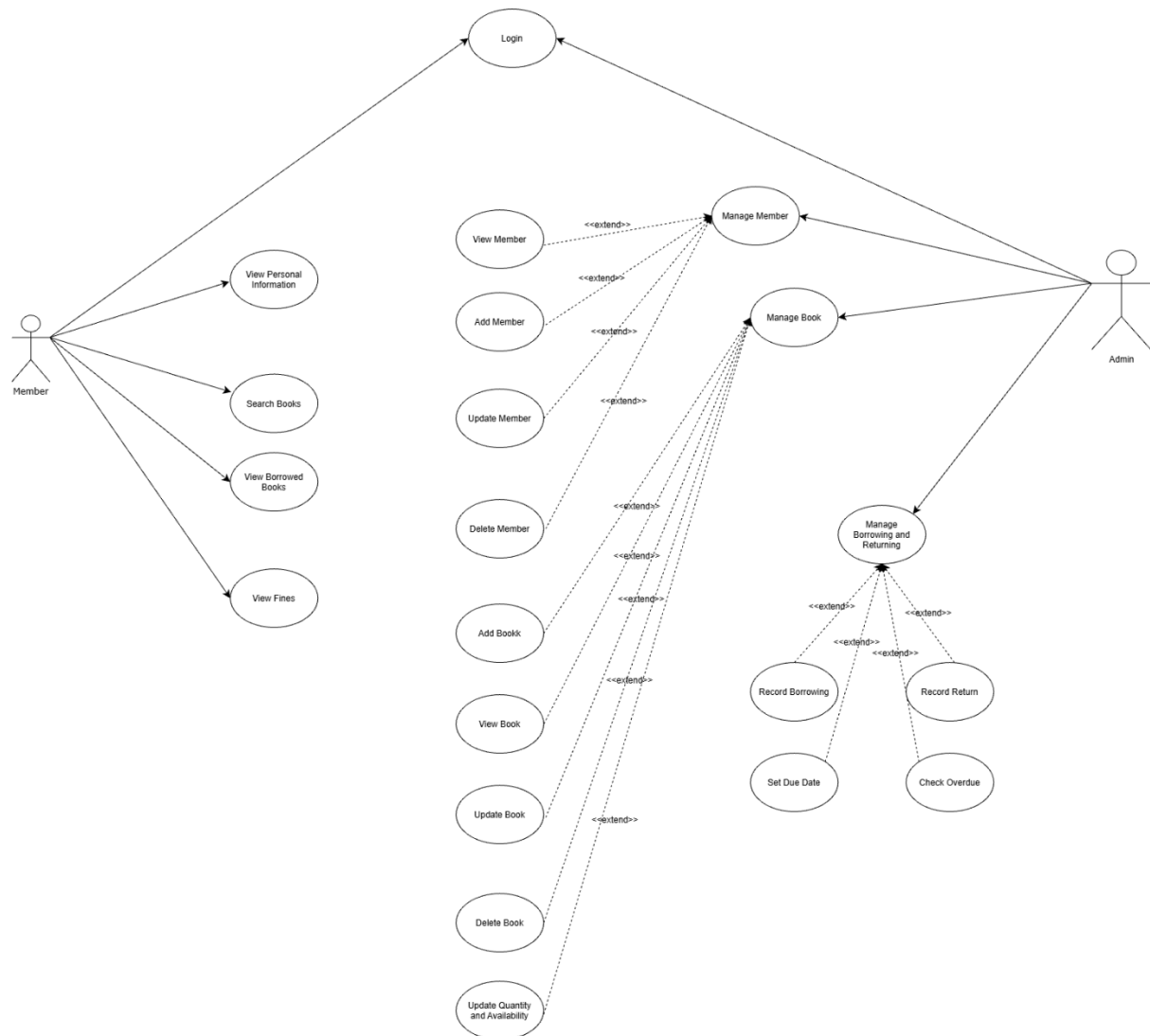


Level 1



III. USE CASE DIAGRAM

1. Overview Diagram



STT	Actor	Meaning
1	Member	The main user who can log in, view personal information, search books, view borrowed books, and view fines.
2	Admin	The system administrator who manages members, books, borrowing, returning, and related library records.

2. List of Use-case

STT	Use-case	Meaning	Notes (group)

1	Login	Log in to the system	Member/Admin
2	View Personal Information	View personal information	Member
3	Search Books	Search library books	Member
4	View Borrowed Books	View currently borrowed books	Member
5	View Fines	View overdue fines	Member
6	Manage Members	Manage library members	Admin
7	View Member	Manage library books	Admin
8	Add Member	Manage borrowing and returning	Admin
9	Update Member	Update member information	Admin
10	Delete Member	Delete member record	Admin
11	Manage Books	Manage book records	Admin
12	Add Book	Add new book	Admin
13	View Book	View book information	Admin
14	Update Book	Update book information	Admin
15	Delete Book	Delete book record	Admin
16	Update Quantity and Availability	Update quantity and availability of books	Admin
17	Manage Borrow and Returning	Manage borrowing and returning transactions	Admin
18	Record Borrowing	Create borrowing record	Admin
19	Record Return	Create returning record	Admin
20	Set Due Date	Set return deadline for borrowed books	Admin
21	Check Overdue	Check overdue borrowing status	Admin

3. Use Case: Login

3.1 Summary

This use case allows a Member or Admin to access the Library Management System by entering valid login credentials.

3.2 Flow of Events

Main Flow

1. User enters username and password.
2. System validates login information.
3. System identifies user role.
4. System grants access to authorized functions.

Alternative Flow

- If username or password is incorrect, the system displays an error message.

3.3 Special Requirements

- Username and password must be valid.
- System must control access based on user role.

3.4 System State Before

User is not logged in.

3.5 System State After

User successfully enters the system.

3.6 Extension Point

None.

4. Use Case: View Personal Information

4.1 Summary

Allows a logged-in member to view their personal information.

4.2 Flow of Events

Main Flow

1. Member selects "View Personal Information".
2. System retrieves member information.
3. System displays personal details.

Alternative Flow

- If information cannot be retrieved, system displays an error.

4.3 Special Requirements

- Member can only view their own information.
- Information is read-only.

4.4 System State Before

Member is logged in.

4.5 System State After

Personal information is displayed.

4.6 Extension Point

None.

5. Use Case: Search Books

5.1 Summary

Allows members to search available books in the library.

5.2 Flow of Events

Main Flow

1. Member enters search keywords.
2. System searches book database.
3. System displays matching books and availability.

Alternative Flow

- If no books are found, system displays notification.

5.3 Special Requirements

Search results must display accurate book information.

5.4 System State Before

Member is logged in.

5.5 System State After

Search results are displayed.

5.6 Extension Point

None.

6. Use Case: View Borrowed Books

6.1 Summary

Allows members to view books currently borrowed.

6.2 Flow

Main Flow

1. Member selects borrowed books.
2. System retrieves borrowing records.
3. System displays borrowed book information and due dates.

Alternative Flow

If no borrowed books exist, system displays a notification.

Special Requirements

Member can only view their own borrowing records.

State Before

Member is logged in.

State After

Borrowed book list is displayed.

7. Use Case: View Fines

7.1 Summary

Allows members to view overdue fines.

Flow

1. Member selects View Fines.
2. System retrieves overdue records.
3. System calculates and displays fines.

Alternative Flow

If no fine exists, system displays notification.

Requirements:

Fine information must be accurate.

State Before:

Member is logged in.

State After:

Fine information is displayed.

8. Use Case: Manage Members

8.1 Summary

Allows Admin to manage library members.

Included Use Cases

Manage Members

<<include>>

- View Member
- Add Member
- Update Member
- Delete Member

Flow

1. Admin selects Manage Members.
2. Admin selects required operation.
3. System executes corresponding function.
4. System updates member records.

State Before

Admin is logged in.

State After

Member information is updated.

9. Use Case: View Member

Summary

Allows Admin to view member information.

Flow

1. Admin selects View Member.
2. System retrieves member data.
3. System displays information.

State Before:

Admin logged in.

State After:

Member information displayed.

10. Use Case: Add Member

Summary

Allows Admin to create a new member record.

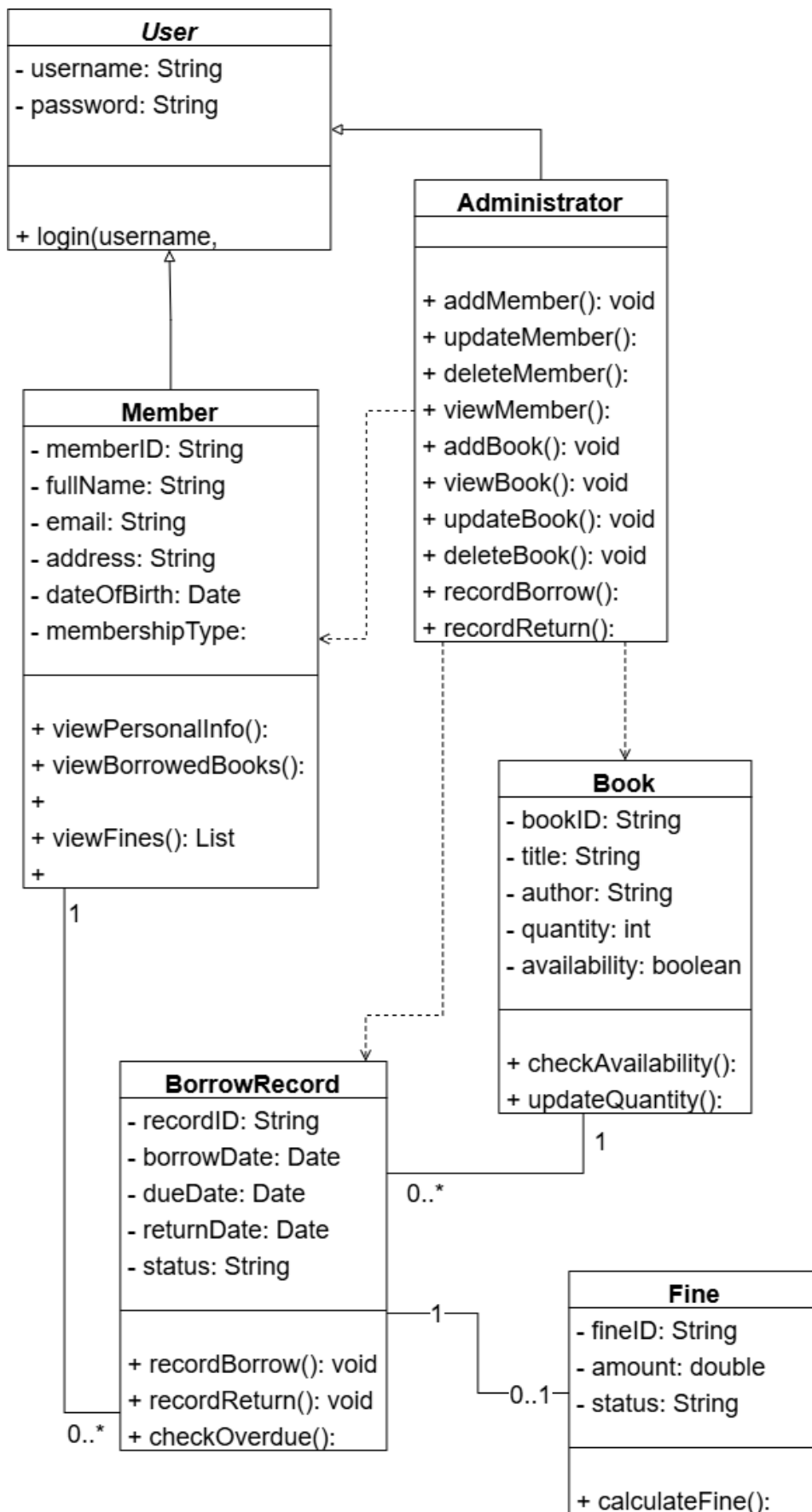
Flow

1. Admin enters member information.
2. System validates data.
3. System saves new member.

Alternative Flow

Invalid information → system displays error.

V. Class Diagram



VI. Data Model

The database consists of four tables: **Members**, **Books**, **BorrowRecords**, and **Fines**. A member can have many borrow records; a book can appear in many borrow records; and each borrow record has at most one associated fine (created only when the book is returned late).

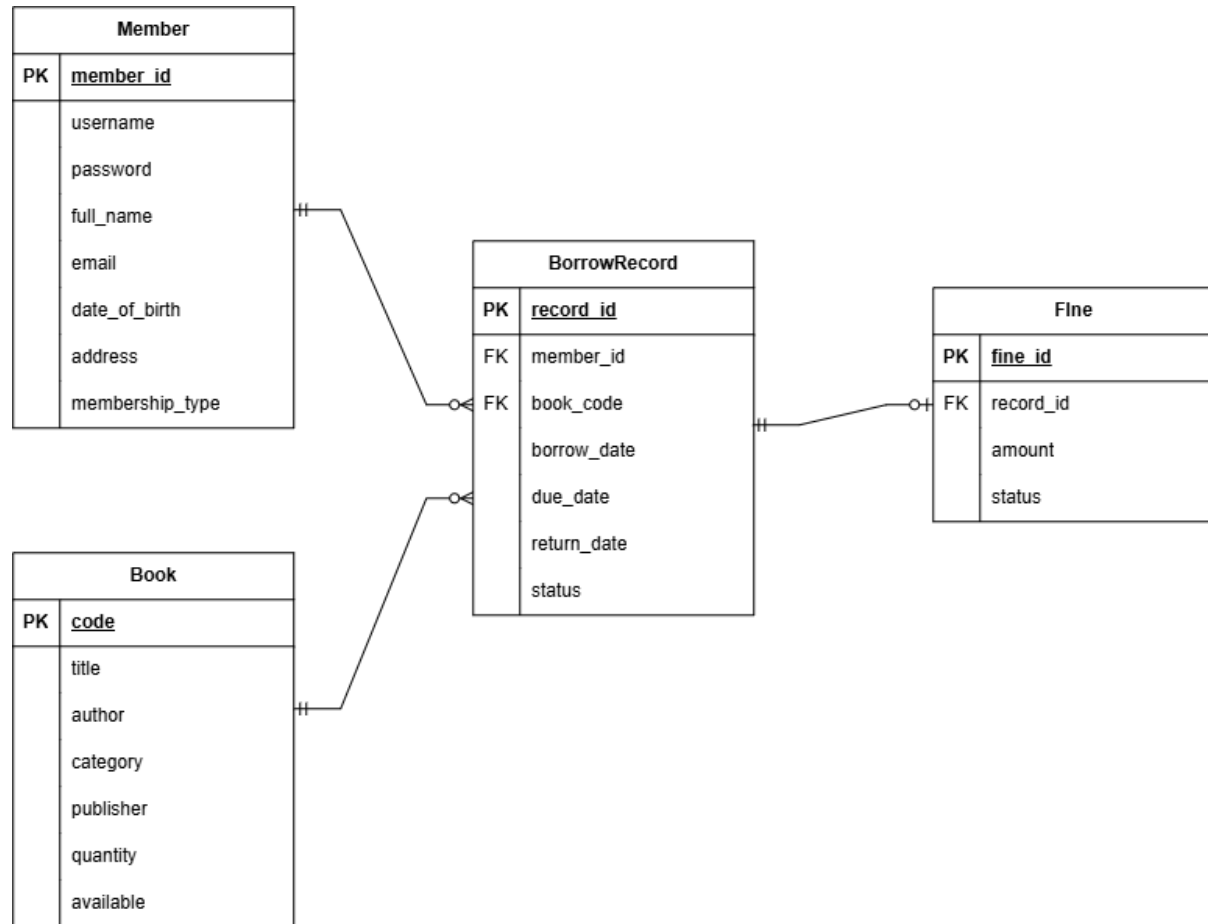


Figure: Entity-Relationship Diagram of the Library Management System database.

Table	Key	Field	Description
Members	PK	member_id	Unique member identifier
		username / password	Login credentials
		full_name, email, date_of_birth, address, membership_type	Personal information shown by “View Personal Information”
Books	PK	code	Unique book identifier
		title, author, category, publisher	Descriptive/searchable fields

		quantity, available	Total copies owned / currently available
BorrowRecords	PK	record_id	Unique borrow-transaction identifier
	FK	member_id → Members	Which member borrowed the book
	FK	book_code → Books	Which book was borrowed
		borrow_date, due_date, return_date, status	Loan lifecycle (Borrowed / Returned / Returned (Overdue))
Fines	PK	fine_id	Unique fine identifier
	FK	record_id → BorrowRecords	Which borrow transaction the fine belongs to
		amount, status	Fine amount (VND) and payment status (Unpaid / Paid)

VII. Interface Design Description

The Library Management System uses a simple console interface. Users interact through a menu and enter the number corresponding to a function.

Login Screen:

Prompt the user to choose a role (member / admin / exit), then enter a username and password.

If the information is incorrect, display “Login failed.” and return to the role prompt.

If login is successful, navigate to the main menu based on the role (Member or Admin).

Member Main Menu:

Display a list of functions:

1. View personal information
2. Search books
3. View currently borrowed books
4. View borrow history
5. View fines
6. Logout

Users select a function by entering the corresponding number.

Admin Main Menu:

Display management functions:

1. Member Management
2. Book Management
3. Borrow / Return Management
4. Logout

Selecting Member Management, Book Management, or Borrow / Return Management opens a further sub-menu:

Member Management: Add member / Edit member / Delete member / View all members / Back

Book Management: Add book / Edit book / Delete book / View all books / Back

Borrow / Return Management: Borrow book / Return book / View all borrow records / View overdue records / View all fines / Pay a fine / Back

Display Screens:

The screenshots above (captured directly from a real run of ``python -m app.main``) demonstrate this:

- Personal information, book search results, borrowed-book lists, borrow history, fines, member lists, book lists, and borrow/fine records are displayed as plain text lines on the console.
- Each list entry is printed on its own line, with fields separated by “|” or “-” so the output stays readable without a graphical table.
- When a list is empty, a clear message is shown instead of a blank screen (e.g. “No books found.”, “You have no books currently borrowed.”, “No outstanding fines.”).

Data Entry Operations:

- Functions for adding, editing, or deleting data require direct input via the keyboard, prompted field by field (e.g. “Member ID:”, “Book code:”, “Due date (YYYY-MM-DD):”).
- The system validates key operations before applying them — e.g. a book cannot be borrowed when its available copies reach 0 (“No copies available.”), and returning a record that does not exist or was already returned is rejected (“Active borrow record not found.”).
- When a book is returned after its due date, the system automatically calculates the number of overdue days and the fine (5,000 VND/day), and creates a Fine record with status “Unpaid”.

GUI Version (Bonus)

In addition to the console interface described above, a graphical desktop version was also implemented using Tkinter (Python's built-in GUI toolkit -- no extra dependency, consistent with the Design Constraint that the system must run directly using Python). This GUI is a visual layer only: every screen calls the exact same business-logic functions used by the console app (auth.py, member_ops.py, admin_ops.py, fines.py) -- none of that logic was duplicated or modified. Both interfaces read and write the same data/sample_data.json and can be used interchangeably.

Run with: `python -m app.gui_main`

Login Screen

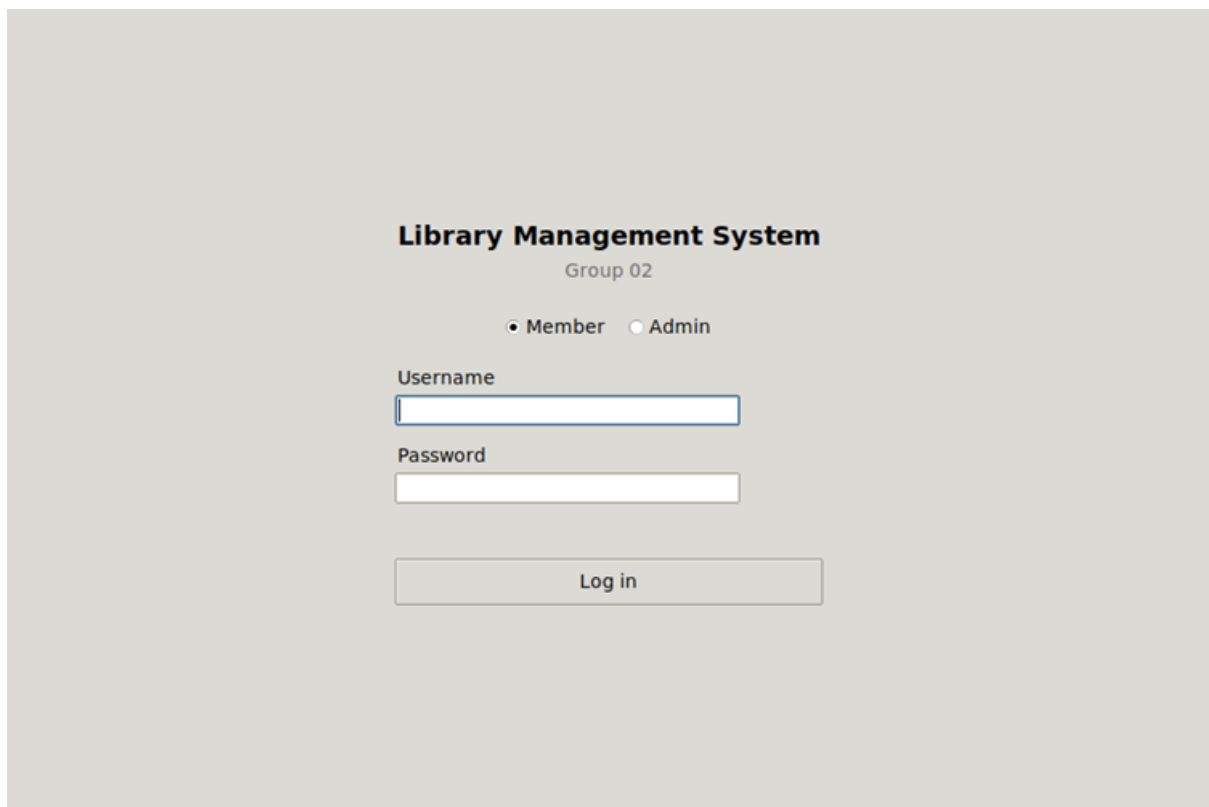
The image shows a graphical login screen for a 'Library Management System'. The title 'Library Management System' is centered at the top in a bold, black font. Below it, 'Group 02' is displayed in a smaller, regular font. There are two radio buttons for role selection: 'Member' (which is selected) and 'Admin'. Below the role selection, there are two text input fields: one for 'Username' and one for 'Password'. At the bottom of the form is a 'Log in' button. The entire interface is set against a light gray background with a subtle grid pattern.

Figure: GUI login screen -- role selection (Member/Admin), username and password.

Member Dashboard

MEMBER MENU		Personal information	
Personal information		Full Name: Nguyen Van A Member ID: LIB001 Email: member01@ut.edu.vn Date of Birth: 2001-05-20 Address: 123 Nguyen Trai, Q5, TP.HCM Membership Type: Standard	
Search books			
Currently borrowed books			
Borrow history			
Fines			
Logout			

Figure: Member dashboard -- Personal information, matching the console's "View personal information" output exactly.

MEMBER MENU		Search books	
Personal information		clean Search	
Search books		Search Results: - BK001 Clean Code Robert C. Martin Available: 2/3	
Currently borrowed books			
Borrow history			
Fines			
Logout			

Figure: Member dashboard -- Search books, showing the same result text the console's `search_books()` function returns.

Admin Dashboard

The screenshot shows the 'Members' section of the Admin Dashboard. On the left is a sidebar with an 'ADMIN MENU' containing 'Members', 'Books', and 'Borrow / Return', and a 'Logout' button at the bottom. The main area is titled 'Members' and includes buttons for 'Add member', 'Edit member', 'Delete member', and 'Refresh'. Below these is a table with the following data:

member_id	username	full_name	email	membership_t
LIB001	member01	Nguyen Van A	member01@ut.κ	Standard

Figure: Admin dashboard -- Members list with Add / Edit / Delete / Refresh actions.

The screenshot shows the 'Borrow / Return Management' section of the Admin Dashboard. The sidebar is identical to the previous screenshot. The main area is titled 'Borrow / Return Management' and includes buttons for 'Borrow book', 'Return book', and 'Pay a f'. Below these are buttons for 'View all records', 'View overdue records', and 'View all fines'. The 'View all fines' button is active. Below these buttons is a table with the following data:

fine_id	record_id	amount	status
FN001	BR002	35000	Unpaid

Figure: Admin dashboard -- Borrow/Return Management, "View all fines" after returning a book late (Fine FN001, 35,000 VND, auto-created and Unpaid).

